

	MSF - OCP	Intensive care
	<u>DIABETIC KETOACIDOSIS (DKA) IN ADULTS</u>	

I – DIAGNOSIS

A/ Clinical presentation

Ketoacidosis can occur in known diabetic patients (type 1 or 2) or may be the initial manifestation of new onset diabetes. It evolves acutely over a maximum of several days.

The symptoms are variable, but generally associate:

- Dehydration (weight loss, decreased skin turgor, tachycardia, even hypotension...)
- Gastrointestinal problems: abdominal pain, nausea/vomiting
- Tachypnoea (respiratory rate >22/min), deep laboured respiration (Kussmaul breathing) with ketone (fruity) smell on the breath
- More rarely, obtundation, stupor...
- Polyuria, polydipsia on history-taking

B/ Definition

DKA associates:

- Hyperglycaemia (> 250 mg/dl or 13.9 mmol/L)
- Ketonuria on urine dipstick $\geq 2+$
- Metabolic acidosis (pH < 7.3 ; HCO_3^- < 18 mmol/L)

In the MSF context, the association of hyperglycaemia and ketonuria with the typical clinical findings is sufficient to make the diagnosis.

	Minor	Moderate	Severe
Hyperglycaemia	> 250 mg/dl (13.9 mmol/L)	> 250 mg/dl (13.9 mmol/L)	> 250 mg/dl (13.9 mmol/L)
Ketonuria	$\geq 2+$	$\geq 2+$	$\geq 2+$
Ketonaemia	3 - 4 mmol/L	> 4 - 8 mmol/L	> 8 mmol/L
pH	7.25 - 7.30	7.00 - 7.24	< 7.00
HCO_3^-	15 - 18	10 - < 15	< 10
Neurological state	Alert	Alert/confused	Obtunded/drowsy

Suspect metabolic acidosis if there is an increased anion gap :

$$(\text{Na}^+) - (\text{Cl}^- + \text{HCO}_3^-) > 12 \text{ mmol/L}$$

C/ Pathophysiology

Reduced plasma insulin levels (relative or absolute reduction) promotes hyperglycaemia and leads to glycogenolysis and neoglucogenesis, which perpetuates the hyperglycaemia. This in turn results in glycosuria with osmotic diuresis, and subsequent dehydration and electrolyte losses. Renal perfusion is subsequently diminished, causing less excretion of glucose, further aggravating the hyperglycaemia.

Lack of insulin also favours lipolysis and formation of ketone bodies from fatty acids, for which the catabolism and renal elimination are reduced. The combination of all of these metabolic abnormalities accounts for the hyperglycaemia, glycosuria, ketonuria and acidosis metabolic seen in DKA.

D/ Additional complementary investigations

According to availability :

- Arterial/venous blood gas (pH)

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- Electrolytes : K⁺, Na⁺, HCO₃⁻, Cl⁻, Mg⁺⁺, phosphate...
- Renal function (urea, creatinine)
- Search for precipitating factor : pregnancy test, full blood count (infection)...
- ECG if abnormal K⁺

II- DIFFERENTIAL DIAGNOSTIC

Ketoacidosis related to fasting/starvation

- Alcoholic ketoacidosis
- Hypersmolar coma

In addition to the clinical history, the following findings can help to distinguish the cause :

	DKA	Ketoacidosis FASTING/ALCOHOL	Hyperosmolar coma
Blood glucose level (BGL)	> 250 mg/dl (13.9 mmol/L)	Normal (hypoglycaemia possible)	> 500 mg/dl (> 28 mmol/L)
Glycosuria	++	-	++
Ketonuria	≥ 2+	1-2+	0-1+
pH	< 7.30	Variable (low or N)	> 7.30
HCO ₃ ⁻	< 18 mmol/L	< 18 mmol/L	> 18 mmol/L
Osmolarity	↑	Normal	↑↑↑
Neurological state	Alert	Alert	confused/comatose

III - TREATMENT

A/ Rehydration

Significant dehydration can develop (osmotic diuresis, GIT losses,...). Intravenous rehydration should be initiated with NaCl 0.9%.

Approximate volume to administer over 24 hours = 5-7L (to be adjusted according to evolution, tolerance, diuresis, patient weight < 50kg,...):

Start by giving a 1L bolus of NaCl 0,9% as quickly as possible, then :

Hour	H1	H2	H3	H4	H5	H6	H7	H8→H13	H14→H21
Volume (ml)	1000	500	500	250	250	250	250	1L sur 6h	1L sur 8h

The patient in shock will need more rapid volume resuscitation (intravascular filling) in the initial phase until peripheral perfusion is adequately corrected (may also require vasopressors).

Once the plasma glucose level falls under 200 mg/dl (11 mmol/L), NaCl 0.9% should be replaced with an IV solution containing sodium and glucose in order to avoid normalising the BGL too rapidly (target for hourly correction of BGL = 3-4 mmol/L/hour, see below) :

Preparation:

- Remove 100 ml from 1L NaCl 0.9%, then add 100 ml of 50% glucose to obtain solution of NaCl/5% glucose
- OR
- Add 4g/L Na⁺ to 1L 5% glucose

If blood glucose levels are too low to permit Insulin therapy (with persisting ketonuria), 10% glucose can be used instead of 5% glucose (200ml 50% glucose + 800ml NaCl 0.9% or Ringer lactate). If

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corrected natremia (see paragraph III-E) is $>150\text{mmol/L}$, Ringer Lactate can be chosen over NaCl 0.9%.

B/ Insulin therapy

Rehydration will have already started to reduce blood glucose levels. If the correction is too rapid, there is a risk of cerebral oedema, so the **objective is to reduce BGL by 3-4 mmol/L/hour. Insulin administration must be continued while ketonuria is still present** (even if this entails lower doses of Insulin). Glucose intake should be increased as necessary to allow the continuation of Insulin.

Insulin therapy is only started after the 1st hour of fluid resuscitation (i.e.. after 2L given : 1L bolus + 1L in H1) AND if the K⁺ plasma level is $> 3.5\text{ mmol/L}$ (if $K^+ \leq 3.5\text{ mmol/L}$, wait for correction of K⁺ plasma level). If impossible to measure K⁺ plasma level, Insulin will be started under cover of supplementary potassium in the IV infusion fluid.

B.1. IV (electric syringe driver)

Rapid-acting Insulin only.

- Bolus 5 IU if BGL is impossible to measure (glucometer shows 'Hi') before commencing the continuous infusion
- Then 0.1 UI/kg/hour as the starting rate

If speed of BGL correction is $< 3\text{ mmol/h} \rightarrow$ increase infusion rate by 2 IU/hour every hour

If speed of BGL correction is $> 4\text{mmol/h} \rightarrow$ decrease infusion rate by 2 IU/hour every hour

Maximum infusion rate = 15 UI/hour

Once the BGL is $\leq 200\text{ mg/dl}$ (11 mmol/L), reduce infusion rate by 50%, and add glucose to the IV infusion fluid (see above). Insulin administration must be continued while ketonuria is still present (if needed, decrease insulin rate and increase glucose).

If the BGL remains 'Hi' (impossible to measure the value), give a further IV bolus of 5 IU every 2 hours.

B.2. Subcutaneous (SC)

Rapid-acting Insulin only.

0.1 UI/kg every hour.

If speed of BGL correction is $< 3\text{ mmol/h} \rightarrow$ increase dose by 0.05 IU/kg every hour

If speed of BGL correction is $> 4\text{mmol/h} \rightarrow$ decrease dose by 0.05 IU/hour every hour

Maximum dose = 15 UI/hour

Once the BGL is $\leq 200\text{ mg/dl}$ (11 mmol/L), reduce dose by 50%, and add glucose to the IV infusion fluid (see above). Insulin administration must be continued while ketonuria is still present (if needed, decrease insulin dose and give it every 2 hours, and increase glucose).

If the BGL remains 'Hi' (impossible to measure the value) before 1st subcutaneous dose, consider concurrent administration of an IV bolus of 5 IU.

C/ Potassium

Total body potassium is generally very depleted.

C.1. Plasma K⁺ can be measured

Measure plasma K⁺ at least every 4 hours.

No potassium supplementation if plasma K⁺ $> 5.3\text{mmol/L}$.

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If plasma $K^+ \leq 5.3 \text{ mmol/L}$, give potassium supplementation by adding 15-20 ml of 10% KCl 10% (20-30 mmol = 1.5-2g) to each litre of IV rehydration fluid (except first litre given as a bolus). More potassium may be required according to repeated plasma K^+ (maximum 40 ml = 52 mmol = 4g per litre to be given over a minimum of 4 hours).

If K^+ is initially very low ($<3 \text{ mmol/L}$), continuous IV administration by electrical syringe can be considered on a second IV access (concurrently to potassium enriched rehydration : 10ml of 10% KCl+ 40ml of saline over 2 hours).

IV potassium administration must be strictly controlled (ideally by an infusion pump) in order to not exceed maximum of 1-1.5g/hour. IV insertion site should be checked regularly and changed if any doubt of extravasation (KCl is toxic for vein and can lead to cutaneous necrosis). If the patient has anuria after 3liters of rehydration, stop KCl administration, check K^+ and renal function tests.

C.2. Plasma K^+ cannot be measured

- ECG is available and shows signs of hyperkalaemia (tall peaked T waves, symmetrical changes...) : wait for urine output $> 0.5 \text{ ml/kg/hour}$. Re-check ECG : if signs of hyperkalaemia have disappeared, give potassium supplementation by adding 15 ml of 10% KCl 10% (20 mmol) to each litre of IV rehydration fluid (except first litre given as a bolus).
- ECG is available and shows no signs of hyperkalaemia : give potassium supplementation by adding 15 ml of 10% KCl 10% (20 mmol) to each litre of IV rehydration fluid (except the first litre given as a bolus). Monitor urine output and stop supplementation if output $< 0.5 \text{ ml/kg/hour}$.
- ECG is not available : wait for urine output $> 0.5 \text{ ml/kg/hour}$, give potassium supplementation by adding 15 ml of 10% KCl 10% (20 mmol) to each litre of IV rehydration fluid (except first litre given as a bolus).

D/ Precipitating factor

Search et treat for any identified precipitating factor (antibiotic therapy for bacterial infection, anti-malarial if malaria,...). Newly-diagnosed diabetes or non-compliance with treatment in established diabetes are often responsible.

E/ Other

Correct other potential electrolyte abnormalities.

Reminder : serum sodium corrected (Na_c) for BGL could make Ringer lactate preferable to NaCl 0.9% :

$Na_c = Na + (0.3 \times (\text{BGL in mmol} - 5))$. Ex : Na^+ measured = 142mmol/L, BGL = 32 mmol/L. $Na_c = 142 + (0.3 \times (32 - 5)) = 142 + 8 = 150 \text{ mmol/L}$
 $Na_c = Na + (1.6 \times (\text{BGL in g/L} - 1))$. Ex : Na^+ measured = 142mmol/L, BGL = 350 mg/dl = 3.5 g/L. $Na_c = 142 + (1.6 \times (3.5 - 1)) = 146 \text{ mmol/L}$

IV – MONITORING

A/ Clinical

Vital signs : BP, HR, SpO_2 , respiratory rate, temperature

Urine output (insert urine catheter)

Neurological monitoring (AVPU, pupils)

Pain

➔ Every hour initially, then with frequency according to evolution

B/ Investigations

BGL (glucometer) / h

Urine dipstick and plasma electrolytes / 4h

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➔ As long as ketonuria is still present

After ketonuria has disappeared (ketones negative) AND electrolyte abnormalities are corrected, perform tests less frequently (BGL/4h, urine dipstick and plasma electrolytes/12h) for the next 24h.

V- RELAY BY CLASSIC INSULIN THERAPY

A/ Re-starting oral fluids/diet

Oral hydration is possible once conscious state permits and BGL can be measured with stable doses of Insulin. Diabetic diet can be resumed when ketonuria has disappeared.

B/ Insulin therapy

Relay by classical Insulin therapy (basal/bolus) is possible once ketonuria has disappeared, oral feeding is well tolerated, and vital signs are normal.

- If daily dose of rapid insulin = X
 - Divide X between slow acting and rapid insulin :
 - $\frac{1}{2}$ X slow acting insulin in the evening
 - $\frac{1}{2}$ X divided over the day in 3 injections of rapid insulin before meals
- If there is no slow insulin, 2/3X of intermediate insulin (NPH) in the morning , 1/3X in the evening + bolus of rapid insulin if needed
- Consider introducing or resuming oral anti diabetic treatment if no contra indication once the acute episode has resolved.